



CS1002 – Programming Fundamentals

Lecture # 05
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Rizwan Ul Haq

Pseudocode – Comparing two numbers

```
1.0 start
2.0 declare n1, n2
3.0 input n1,n2
4.0 if (n1 > n2)
    4.1 print "n1 is greater"
5.0 endif
6.0 if (n2 < n1)
    6.1 print "n2 is greater"
7.0 endif
8.0 if (n1 = n2)
    8.1 print "they are equal"
9.0 endif
10.0 end
```

Nested **if/else** Selection Structure

- Nested if/else structures
 - One inside another, test for multiple cases
 - Once condition met, other statements skipped

Example 1– Comparing two numbers with nested if-else

1.0 start

2.0 declare n1, n2

3.0 input n1,n2

4.0 if (n1 > n2)

 4.1 print "n1 is greater"

5.0 else

 5.1 if (n2 < n1)

 5.1.1 print "n2 is greater"

 5.2 else

 5.2.1 print "they are equal"

 5.3 endif

6.0 endif

7.0 end

Example 2: Grading with Nested if-else:

```
if student's grade is greater than or equal to 90
    Print "A"
```

```
else
```

```
    if student's grade is greater than or equal to 80
        Print "B"
```

```
    else
```

```
        if student's grade is greater than or equal to 70
            Print "C"
```

```
        else
```

```
            if student's grade is greater than or equal to 60
                Print "D"
```

```
            else
```

```
                Print "F"
```

Example 2: Pseudocode for Grading with Nested if-else:

```
if ( grade >= 90 ) then           // 90 and above
    Print "A"
else if ( grade >= 80 ) then      // 80-89
    Print "B"
else if ( grade >= 70 ) then      // 70-79
    Print "<< "C";
else if ( grade >= 60 ) then      // 60-69
    Print "D";
else                             // less than 60
    Print "F";
end
```

if/else Selection Structure

Compound statement

- ➡ Set of statements within a block

```
if  grade >= 60 then
    Print "Passed"
else
    Print "Failed"
    Print "You must take this course again"
endif
```

Example 3

```
1.0 Declare num1
2.0 Print "Enter a number"
3.0. input num1
4.0  if num1 > 0 then
    4.1 print "You entered a positive number", num1
5.0 else if num1 < 0 then
    5.1 Print "You entered a negative number" , num1
6.0 else
    6.1 print "You entered 0"
7.0 endif
8.0 print "This line is always printed"
```

Example 4

- Write pseudocode and draw a flowchart that will calculate the roots of a quadratic equation
- $ax^2 + bx + c = 0$
- Hint $d = b^2 - 4ac$ and roots are
 - $X_1 = (-b + \text{sqrt}(d)) / 2a$
 - $X_2 = (-b - \text{sqrt}(d)) / 2a$
- Real roots are only possible if $d > 0$
- **Cater this as well**

Example 5

- Write a pseudocode and draw a flowchart
 - to read an employee name (NAME), overtime hours worked (OVERTIME), hours absent (ABSENT)
 - determine the bonus payment (PAYMENT)

Bonus Schedule	
$\text{OVERTIME} - (2/3) * \text{ABSENT}$	Bonus Paid
>40 hours	5000
>30 but $\leq$ 40 hours	4000
>20 but $\leq$ 30 hours	3000
>10 but $\leq$ 20 hours	2000
$\leq$ 10 hours	1000

Questions

